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## CS 306: Operating Systems

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| Course Code | CS 306                |
| Course Name | Operating Systems     |
| L-T-P-C     | 3-0-0-3               |
| Approval    | 5th Senate (Oct 2013) |

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### Course Contents

- **Introduction:** Review of Computer Organization and Architecture, OS Concepts, Structure and Operations - Process Mgmt, Memory Mgmt. and Storage Mgmt.
- **Process Management:** Processes - Concept, Scheduling - different scheduling algorithms, Need for synchronization, Process Synchronization - Critical Section, Mutexes, Semaphores, Mailboxes, Monitors, Inter-process Communication, Deadlock and its prevention.
- **Memory Management:** Physical memory vs Virtual memory, Swapping, Paging and Page tables, Segmentation, Address translation and MMU, Caches, Page misses and TLB, Handling page miss and Page replacement.
- **File Systems and I/O Management:** Concept of File and File System, Structure of FS, Allocation and management of Free space, brief review of mass storage structure and its relation to FS implementation. Generic I/O interface presented by OS for FS, Network - Device handling.
- **Module V:** Protection and Security as applicable to Process, Memory and File Management. Virtualization. Case study of an OS (including details of data structures and algorithms at the kernel level).